Rooftop Solar Screening from OpenStreetMap Building Footprints

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***Abstract*—Before paying for a solar design study, a building owner needs a quick first answer: how much solar power fits on the roof, and does it pay back? In the United Arab Emirates, most automated tools cannot give this answer, because roof laser scans and 3D city models are not public. We present SolarScan, an open-source tool that uses free data only. It takes an address, finds the building outline in OpenStreetMap, lays out the solar modules on the roof, and estimates the yearly energy and the payback time. To test it, we traced the roofs of 24 buildings in 5 emirates by hand. When the map outline matched the building (17 of 24), the roof area was within 4.4% of the trace on average. The other 7 outlines were wrong: part of the building was missing, open ground or a courtyard was counted as roof, or the roof had several levels. For a university building, the tool estimates 172.4 kWp, 293.9 MWh per year and a payback of 4.6 years. Cost and electricity price change the payback most; the panel direction changes it by only 0.3%.**

***Index Terms*—building footprint, open-source software, OpenStreetMap, photovoltaic systems, PVGIS, rooftop solar**

I. Introduction

Rooftop photovoltaic (PV) systems suit the United Arab Emirates (UAE) well: at the University of Sharjah, each square meter of flat ground receives about 2,191 kWh of sunlight per year. Yet many projects stop at the first question: *is this roof worth a closer look?* To answer it, an installer usually visits the site. This takes time and money, and an owner of many buildings, such as a school group or a logistics company, needs the answer for all of them at once.

Automated tools exist, but they need data that the UAE does not have. Large studies use airborne light detection and ranging (LiDAR) or 3D city models [1], [2]. Others use land-registry data with machine learning [3]. Commercial services use licensed images and charge per request. None of these gives a free first answer in the Gulf, although rooftop PV can be economically viable in the UAE [4], [5].

OpenStreetMap (OSM) is a free world map made by volunteers. It contains building outlines [6], but their quality varies from place to place [7]–[10]. The Photovoltaic Geographical Information System (PVGIS) of the European Commission gives free solar energy estimates for any point in Europe, Africa and the Middle East [11]. Together, the two sources give a roof and the energy it can produce.

We built a tool on this idea and asked three questions:

- **Q1:** How close is the OSM roof area to the real roof?
- **Q2:** What does a full screening report look like for a real building?
- **Q3:** Which inputs change the answer most?

The paper makes three contributions. First, SolarScan, an open-source tool that goes from an address to a report using only OSM and PVGIS. Second, a public test of OSM roof areas on 24 UAE buildings, with checked hand traces of the roofs and a description of how OSM outlines fail. Third, a sensitivity study that shows which inputs matter.

TABLE I
Related Work by Main Data Source

Work	Main data	Region	One address
Gagnon *et al.* [2], [19]	LiDAR and GIS	USA	No
Walch *et al.* [1]	3D city model	Switzerland	No
Assouline *et al.* [3]	Land registry	Switzerland	No
Mainzer *et al.* [16]	OSM and statistics	Germany	No
Mainzer *et al.* [17]	OSM and images	Germany	No
Joshi *et al.* [18]	Building footprints	World	No
Rees *et al.* [20]	LiDAR and open data	Norway	No
Yu *et al.* [14]	Aerial images	USA	No
pvlib, PVGIS [11], [15]	Roof given by user	World	Yes
This work	OSM and PVGIS	UAE	Yes

II. Related Work

Table I groups earlier work by its main data source. Reviews describe many methods that use geographic information systems (GIS) to estimate rooftop PV potential [12], [13]. Most of them need LiDAR, 3D building models or city data. Deep-learning methods find roofs or panels in aerial images [14], but they need large image sets and strong computers. Energy models such as pvlib [15] and PVGIS [11] are accurate, but the user must describe the roof. Mainzer *et al.* [16] used OSM to estimate the solar potential of a whole country, and later combined OSM outlines with aerial images to find roof planes in a city [17]. Joshi *et al.* [18] mapped rooftop PV worldwide from building footprints and machine learning; they name the footprints as the largest source of error. To our knowledge, no open tool turns one address into a screening report, and no study has checked OSM roof areas for this use in the UAE.

III. Method

SolarScan works in six steps: find the building, measure the roof, remove the edge strip, lay out the modules, estimate the energy and estimate the payback. Fig. 1 shows the result for one building. Table II lists the default values and their sources. The user can change every value in a settings file.

A. Find the Building

The user gives an address or coordinates. The tool asks OSM for all buildings within 250 m. It selects the building that contains the point or, if there is none, the closest one. If OSM has no building there, the tool stops with an error message. It never invents an outline.

B. Measure the Roof

OSM stores each corner of the outline as latitude ϕ and longitude λ. We convert them to meters on a flat plane around the center of the building (ϕ_0, λ_0):

$$x_i = N \cos\phi_0 \, (\lambda_i - \lambda_0), \qquad y_i = M \, (\phi_i - \phi_0), \tag{1}$$

where M and N are the radii of curvature of the Earth at ϕ_0 in the World Geodetic System 1984. The roof area A and the perimeter P follow from the corners (x_i, y_i):

$$A = \frac{1}{2} \Big| \sum_{i=1}^{n} (x_i y_{i+1} - x_{i+1} y_i) \Big|, \qquad P = \sum_{i=1}^{n} d_i, \tag{2}$$

where d_i is the length of side i and corner $n+1$ is corner 1. For all test buildings, this area is within 0.0004% of the exact area on the Earth's surface.

C. Find the Usable Area

Safety rules ask for a clear strip along the roof edge. We use a strip of width $s = 1.5$ m and shrink the outline inward by s to get the usable area A_u. For a simple roof, this gives about

$$A_u \approx A - sP. \tag{3}$$

So a small or narrow building, with a large perimeter for its area, loses more of its roof.

D. Lay Out the Modules

On a flat roof, the modules stand in tilted rows that face south, as in [19]. Each module is 1 m wide and $L = 2$ m long, and is tilted by 15°. The rows need gaps so that one row does not shade the next. We space them by L/k, where k is the share of roof covered by modules. With $k = 0.70$, the rows are 2.86 m apart. At noon on 21 December, the sun is at its lowest of the year (41°). The shadow of a row then reaches 0.59 m, less than the 0.93 m gap to the next row (Fig. 1d). On that day, the rows do not shade each other from 08:46 to 15:46. PVGIS does not model the shading outside these hours. The tool places a module only where its whole footprint fits inside the usable area, and counts the modules N (Fig. 1b). The direct-current (dc) capacity is

$$P_{\mathrm{dc}} = N \, P_{\mathrm{mod}}, \tag{4}$$

TABLE II
Assumed System and Default Values

Item	Default	Source
What is installed		
Module	400 W, 1 m × 2 m	typical
Module type	crystalline silicon, 20%	typical
Mounting	fixed racks, 15°, facing south	[19]
Row spacing	2.86 m pitch ($k = 0.70$)	[19]
Edge strip s	1.5 m	assumption
Inverter	dc/ac ratio 1.2	[19]
Grid	connected, no battery	assumption
How the energy is computed		
Solar data	PVGIS 5.3, satellite	[11]
System losses	14% (cables, inverter, dirt)	PVGIS
Module ageing	0.5% per year	assumption
How the money is counted		
Installed cost c	3,000 AED/kWp	installers
Electricity price r	0.38 AED/kWh	top tariff band
Energy use	all used in the building	assumption
Maintenance	1.5% of cost per year	assumption
Discount rate, lifetime	5%, 25 years	assumption

where $P_{\mathrm{mod}} = 400$ W is the rating of one module with an efficiency of 20%. A simpler rule, $P_{\mathrm{dc}} = k\,\eta\,A_u \cdot 1$ kW/m^2, ignores that whole modules do not fill the roof edges. We compare the two in Section V-B. An inverter converts the dc power to alternating current (ac); we size it at the dc capacity divided by 1.2.

E. Estimate the Energy

PVGIS uses satellite data and models the effects of heat and sun angle on the panels [11]. We downloaded its yearly energy per kilowatt of rated module power, Y (kWh per kWp, kilowatt-peak), for all tilts and directions at one site in each emirate. The tool reads Y from these stored tables, so it works offline. The yearly energy is

$$E = P_{\mathrm{dc}} \, Y. \tag{5}$$

To check PVGIS, we modeled the same system with pvlib [15], which uses its own weather and panel models. The two tools agreed within 1.8% for seven tilts and directions.

F. Estimate the Payback

With an installed cost c per kWp and an electricity price r, the simple payback in years is

$$T = \frac{c \, P_{\mathrm{dc}}}{r \, E} = \frac{c}{r \, Y}. \tag{6}$$

The payback does not depend on the roof size. It depends only on cost, price and yield. The tool also reports the net present value (NPV) and the levelized cost of electricity (LCOE) over 25 years. All money values are in UAE dirham (AED).

IV. Test Setup

A. Buildings

We chose 24 flat-roof buildings in 5 emirates: 4 university buildings, 9 shops and malls, and 11 warehouses and workshops (Table III). Their OSM areas range from 124 to 201,710 m^2.

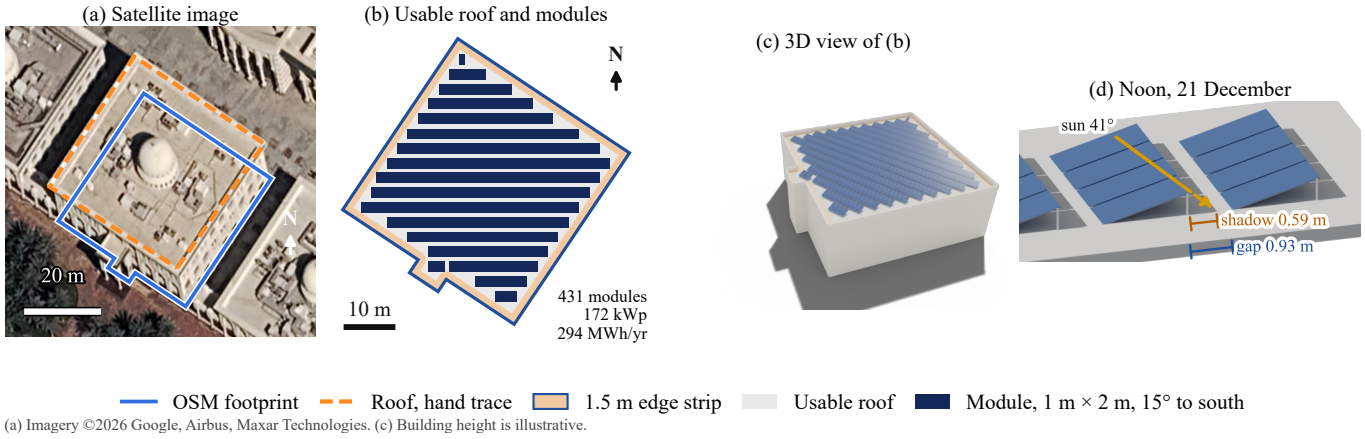


Fig. 1. SolarScan on Building W5, University of Sharjah. (a) Satellite image with the OSM outline and the hand trace; the shift comes from image misalignment and building lean. (b) Usable roof and module rows. (c) 3D view of (b). (d) Row spacing in the noon sun of 21 December.

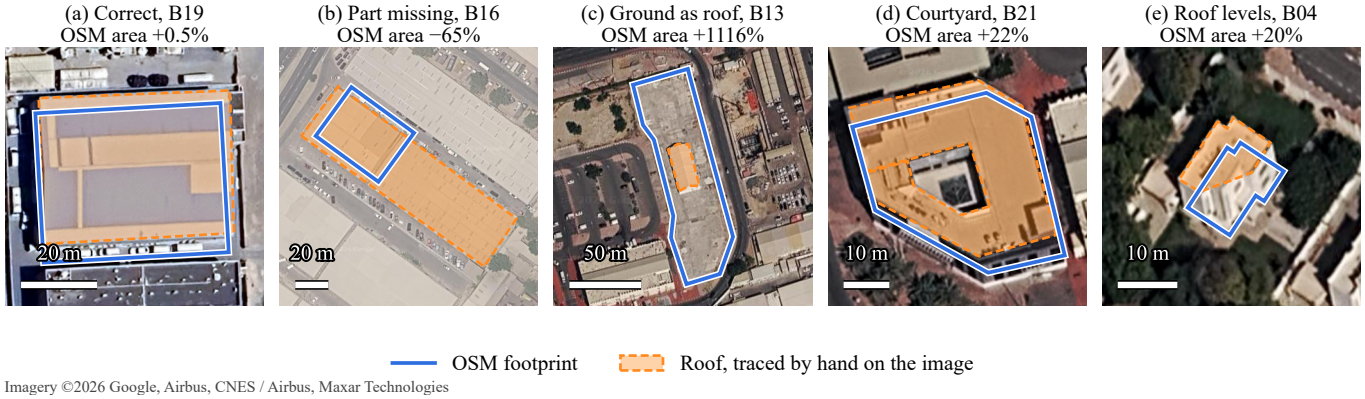


Fig. 2. OSM outline (blue) and hand-traced roof (orange): one correct outline and the four error types. (b) Part of the hall missing. (c) Market ground counted as roof; only the shed has a roof. (d) Courtyard counted as roof. (e) Several roof levels. IDs as in Table III.

We saved each OSM outline, so the results do not change when the map is edited.

B. Reference Roof Areas

No official roof data are public for these buildings, so one author traced every roof by hand on the satellite images in Google Earth. The tracing files showed only a pin at each building, not the OSM outline. Two traces were redrawn after a first check. We then drew every trace and its OSM outline on a satellite image at a known scale (whole Web-Mercator zoom levels only) and checked that the trace followed the roof edge. All 24 traces did. After the comparison, the author looked at the images again and labeled each OSM outline as correct or wrong. A wrong outline got an error type (Table IV). All traces and labels are in the repository.

C. Error Measures

For each building, the area error is

$$e = \frac{A_{\mathrm{OSM}} - A_{\mathrm{trace}}}{A_{\mathrm{trace}}}. \tag{7}$$

We report the mean absolute percentage error (MAPE), the median absolute error and the mean bias error (MBE), which is the mean of e. We do not report R^2, because the areas range over three orders of magnitude and R^2 would be close to 1 for almost any method.

V. Results

A. Q1: Is the OSM Roof Area Correct?

Fig. 2 shows a correct outline and one example of each error type. Fig. 3 shows the error for all buildings, and Table III lists the values.

When the outline is correct. This was the case for 17 of the 24 buildings. Their MAPE was 4.4%, the median error 4.2% and the MBE −1.4%. All errors were between −8.3% and +8.6%, and 10 buildings were within 5%. We found no trend with building size or type.

When the outline is wrong. This was the case for 7 of the 24 buildings (29%), in four different ways (Table IV, Fig. 2). In 4 buildings, OSM mapped only one part of the building, so the area was 20% to 84% too small. In one market, OSM outlined the whole market ground, but only a small shed has a roof. In one office building, the open courtyard was counted as roof. One small building has roofs at several levels, and the

TABLE III
BENCHMARK BUILDINGS: OSM FOOTPRINT AREA AGAINST THE MANUAL ROOF TRACE

ID	Building	Emirate	Type	OSM area (m^2)	Traced area (m^2)	Error (%)	OSM outline
B01	University Building W5 (Computer Science)	Sharjah	Academic	1,607	1,591	+1.0	correct
B02	University Building M4	Sharjah	Academic	1,608	1,550	+3.8	correct
B03	University Building M11	Sharjah	Academic	3,595	3,848	−6.6	correct
B04	University annex building	Sharjah	Academic	124	103	+20.4	several levels
B05	Unnamed building at Mall of the Emirates site	Dubai	Commercial	201,710	209,955	−3.9	correct
B06	Abu Dhabi Mall	Abu Dhabi	Commercial	36,077	34,336	+5.1	correct
B07	City Centre Ajman	Ajman	Commercial	65,533	68,417	−4.2	correct
B08	Unnamed building, Khuzam	Ras Al Khaimah	Commercial	934	992	−5.9	correct
B09	ACE Hardware Al Quoz	Dubai	Commercial	1,950	2,448	−20.3	part missing
B10	Emirates NBD branch building	Dubai	Commercial	3,261	3,313	−1.6	correct
B11	AlSafeer Mall Mussafah	Abu Dhabi	Commercial	2,834	2,661	+6.5	correct
B12	Salem Shopping Center	Ajman	Commercial	3,583	3,907	−8.3	correct
B13	Ajman Public Market	Ajman	Commercial	5,300	436	+1115.9	ground as roof
B14	Warehouse, Jebel Ali (Inchcape Shipping)	Dubai	Industrial	3,935	4,095	−3.9	correct
B15	Alserkal Avenue warehouse	Dubai	Industrial	3,526	3,717	−5.1	correct
B16	On Demand Auto Repairs workshop	Dubai	Industrial	1,701	4,838	−64.8	part missing
B17	TCTI warehouse	Dubai	Industrial	4,641	4,766	−2.6	correct
B18	Sku Line workshop, Mussafah	Abu Dhabi	Industrial	1,716	2,513	−31.7	part missing
B19	Unnamed warehouse, Mussafah M-15 (1)	Abu Dhabi	Industrial	1,904	1,895	+0.5	correct
B20	Unnamed warehouse, Mussafah M-15 (2)	Abu Dhabi	Industrial	1,822	1,877	−3.0	correct
B21	Emirates Post office, Al Jerf	Ajman	Industrial	1,219	1,000	+21.9	courtyard as roof
B22	Unnamed industrial building, Ind. Area 13	Sharjah	Industrial	2,500	2,618	−4.5	correct
B23	Unnamed industrial building, Ind. Area 11	Sharjah	Industrial	1,823	1,678	+8.6	correct
B24	Unnamed industrial building, Ind. Area 10	Sharjah	Industrial	1,514	9,385	−83.9	part missing

Error $= (A_{\mathrm{OSM}} - A_{\mathrm{trace}})/A_{\mathrm{trace}}$. Grey rows: the OSM outline does not match the building (Table IV).

TABLE IV
HOW THE OSM OUTLINE FAILED

Failure type	n	Area error (%)
Part of building missing	4	−84 to −20
Ground area mapped as roof	1	+1116
Open courtyard counted as roof	1	+22
Roof on several levels	1	+20

hand trace covers only the upper one. Over all 24 buildings, the median error was 5.5%; the mean was 59.8% because of the market.

Second reference. We also compared the hand traces with building outlines that Microsoft extracted from satellite images by machine learning [21]. For the 16 buildings with a clear match, their median error was 7.9%, worse than OSM. Like our traces, they were often a few meters away from the OSM outline (Fig. 1a), which changes the area little.

The answer to Q1 is therefore: when the outline is right, the OSM area is within a few percent, but some outlines are wrong. A user should look at the outline before trusting the result.

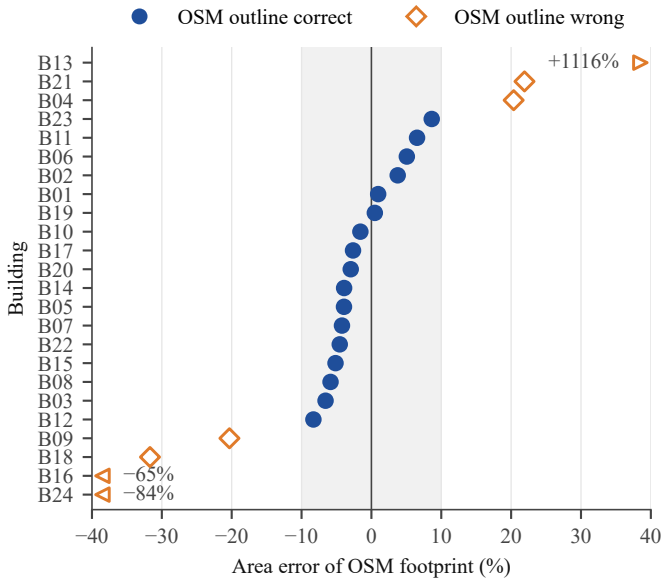


Fig. 3. OSM area error per building, sorted. Gray band: ±10%. Arrows: errors outside the plot.

B. Q2: What Does a Report Look Like?

Fig. 1 shows the result for Building W5 at the University of Sharjah. Its OSM outline has 13 corners and an area of 1,606.8 m^2, +1.0% from the hand trace. After the 1.5 m edge strip, 1,368.9 m^2 (85.2%) is usable. 431 modules fit in south-facing rows, which gives 172.4 kWp. With 1,705 kWh/kWp from PVGIS, the system produces 293.9 MWh per year. At 3,000 AED/kWp and 0.38 AED/kWh, it costs 517,200 AED, saves 111,697 AED per year and pays back in 4.6 years (6.0 years when discounted). Over 25 years, the NPV is 875,235 AED and the LCOE is 0.159 AED/kWh. The user runs the same steps in a desktop interface and gets a one-page report (Fig. 4, shown for a warehouse in Dubai).

Real layout against the simple rule. The simple rule $k\,A_u$ would allow 479 modules on W5, 10% more than the layout. The rows run east to west, while the walls of W5 do not, so

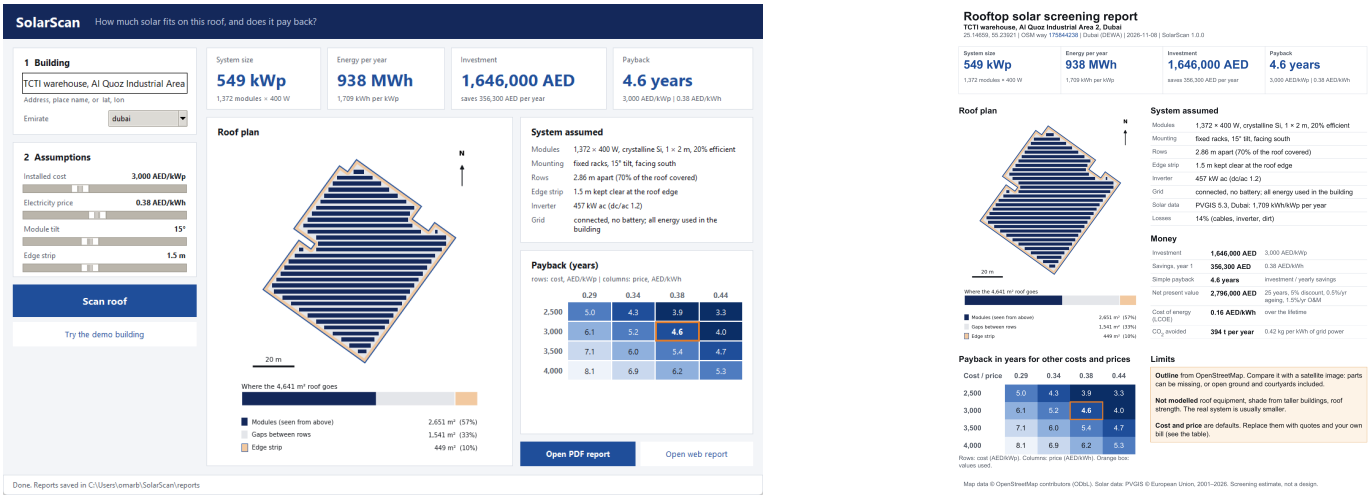


Fig. 4. SolarScan on a warehouse in Dubai (B17). Left: desktop interface; the cost and price sliders update the payback at once. Right: one-page report.

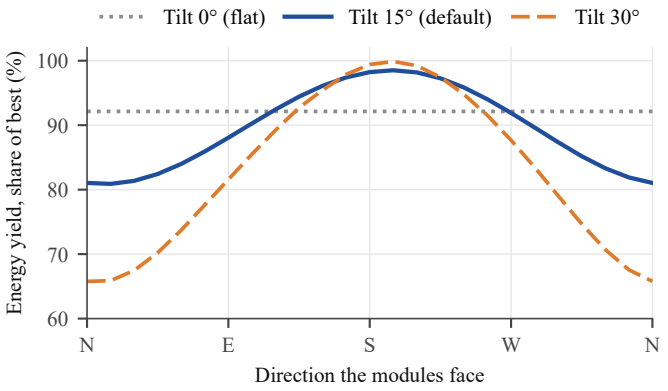


Fig. 5. PVGIS yield in Sharjah by panel direction for three tilts, as a share of the best.

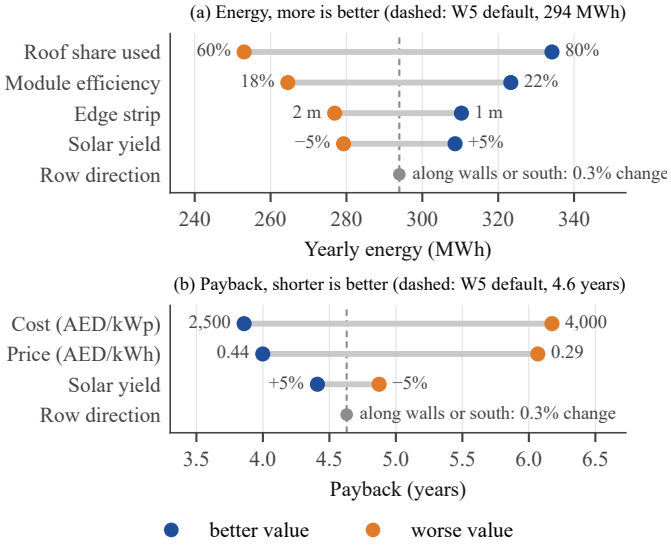


Fig. 6. Effect of each input on Building W5, changed one at a time between the two values shown. Orange: worse value. Blue: better value. Dashed: default.

the triangles at the roof edges stay empty (Fig. 1b). Over all 24 buildings, the layout fits a median of 92% of the modules given by the simple rule (range 64–99%). The lowest value belongs to the smallest roof. The simple rule therefore overestimates the system size by a few percent on large roofs and by much more on small ones. Such constant factors are common, but they ignore the shape of each roof and are hard to validate [22].

C. Q3: Which Inputs Change the Answer?

Panel direction. Fig. 5 shows the PVGIS yield for all directions in Sharjah. At 15° tilt, south gives 98.2% of the best yield, east 88.1%, west 91.9% and north 81.0%. Flat panels give 92.1% in every direction. The sun is high in the UAE, so at low tilt the direction matters little. For W5, rows that follow the walls instead of facing south lose only 0.3% of the energy. Across the five emirates, the yield at 15° south ranges from 1,668 to 1,722 kWh/kWp.

Edge strip. The usable share of the roof falls with the ratio P/A, as (3) predicts. It ranges from 50.1% for the smallest building to 98.3% for the largest. The simple formula (3) differs from the exact result by a median of 0.45 percentage points.

All inputs. Fig. 6 changes one input at a time for W5. The energy depends most on the share of roof covered by modules (the row spacing) and the module efficiency. These are design choices, not errors in the data. The payback depends only on cost, price and yield (6), and ranges from 3.9 to 6.2 years. A 5% error in roof area changes the energy by about 5% and does not change the payback.

VI. Discussion

Use. SolarScan suits a first screen of many flat roofs, such as a campus, a group of warehouses or a city's public buildings. It is not a design tool. It estimates the technical potential; what is actually built also depends on owners, rules and the grid

[23]. Because the payback does not depend on roof size, an area error changes the size of the opportunity, but rarely the decision to look closer.

Limitations. First, OSM gives the outline of the building, not the roof. The satellite image of W5 shows a dome and air-conditioning units on the roof (Fig. 1a). The layout ignores them, as well as skylights, shade from tall neighbors and roof strength, so the real system will be smaller. Second, one person made all traces and labels, after seeing the OSM areas for some buildings. A second, independent tracer would make it stronger. Third, the tool cannot yet detect a wrong outline by itself. Fourth, cost and electricity prices differ between emirates and customers, so users should enter their own values. The default price leaves out the fuel surcharge and value-added tax, so the savings are on the low side. Fifth, PVGIS uses a fixed 14% loss, and dust in the Gulf can cause larger losses when panels are not cleaned [24].

VII. Conclusion

SolarScan gives a free and repeatable first answer to the rooftop solar question for any building in OpenStreetMap. On 24 UAE buildings with hand traces, when the OSM outline was correct (17 buildings), its area was within 4.4% of the trace on average. The other 7 outlines were wrong in different ways. For a typical university building, the tool estimates a 172.4 kWp system with a payback of 3.9 to 6.2 years. Cost and electricity price drive this result, not roof shape or panel direction. Next, we will detect wrong outlines automatically, find roof equipment in open images and test the tool in other Gulf cities.

Data Availability

The code, building outlines, hand traces, PVGIS tables and the scripts that rebuild every number, table and figure in this paper are available at https://github.com/IamOumarIbrahim/rooftop-scanning. The satellite images in Figs. 1 and 2 are © Google and its imagery providers. They are not redistributed, but the script that fetches them is included.

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